Lowe's Store-Level Sales Target Model

ML-Powered Forecasting to Replace Top-Down Allocation

IIM Calcutta Capstone — Group 8 (APAL02)

Live App: https://storewisetargets.streamlit.app/

The Problem: Current Approach

Current Process

- Peanut butter spread: Uses 3-year historical sales share to allocate targets
- One-size-fits-all: Ignores local market dynamics, demographics, and competition
- Manual overrides: Planners adjust targets subjectively with limited data backing
- Inaccurate results: 25.83% of store-weeks miss targets by >±10%

Impact

25.83% plan miss rate

Our Solution: Supervised ML Model

Input	Process	Output
• Store-level data	• XGBoost model	• Accurate predictions
• Demographics	• 40 engineered features	• Under/over-targeted flags
• Competition	• Rolling time split	• 5-segment stratification
• Historical sales	• Store accuracy loop	• Quarterly re-scoring

Key Benefit

Data-driven targeting enables targeted planner overrides instead of chain-wide adjustments

Dataset Overview

269,412	1,727	194	0
Records	**Stores**	**Columns**	**Missing**
store-weeks	unique locations	features	fully clean

Time Period: FY2023–FY2025 (3 fiscal years, 52 weeks each)

Feature Composition

- 14 columns: Identity, time, target (Actual Sales USD)
- 72 columns: Demographics (population, income, housing, education)
- 108 columns: Competition (sister stores, competitor counts by radius)

Phase 1 Feature Engineering: 35 Features in 5 Families

Calendar	2	Year, Fiscal Week
Engineered Lags	6	lag_1, lag_52, roll_4, roll_13
Store/Market	5	Sales Floor Size, Urbanicity, CBSA Type
Demand/ Housing	16	Households, Income, Housing Age, Population
Competition	6	Sister Stores, Competitor Counts, total_competitor_ta

The Power Features: Engineered Lag Features

Built with group-by-store and shift() — no leakage. Strongest predictors of store sales.

lag_1	`shift(1)`	**0.982**
lag_4	`shift(4)`	**0.964**
lag_13	`shift(13)`	**0.913**
lag_52	`shift(52)`	**0.953**
roll_4	`shift(1).rolling(4).mean()`	**0.981**
roll_13	`shift(1).rolling(13).mean()`	**0.964**

Why lags work: Recent sales are the strongest predictor of next period sales

Model Architecture & Configuration

Primary Models

- XGBoost (n_estimators=500, lr=0.05, max_depth=7)
- LightGBM (n_estimators=500, lr=0.05, max_depth=8, num_leaves=63)
- Ridge Regression (regularization baseline)
- Naive Lag-52 baseline (year-over-year forecast)

Training Setup

- Rolling time split: 85th percentile cutoff (temporal integrity)
- Validation set: 50 early stopping rounds
- Hyperparameter tuning: Bayesian optimization on validation WAPE
- Final holdout: Latest ~15% of fiscal weeks (FY2025 Q3–Q4)

Success Criteria

Model Performance Results

Model	WAPE	Bias %	R^2	RMSE
XGBoost	**7.2%**	-0.8%	0.912	8,340
LightGBM	7.4%	+0.3%	0.908	8,520
Ridge	8.8%	+1.2%	0.881	9,100
Lag-52 Naive	8.9%	-0.5%	0.875	9,200

Verdict

XGBoost achieves 7.2% WAPE, beating the success criterion (< 8%) and the naive baseline (8.9%). Model explains 91.2% of variance with minimal bias.

Store-Level Accuracy Loop: Classification

Under-Targeted

bias < -5%

Model predicts less than actual

Well-Calibrated

bias ±5%

Predictions align with actuals

Over-Targeted

bias > +5%

Model predicts more than actual

Per-Store Metrics Computed

- WAPE (Weighted Absolute Percentage Error)
- Bias % (directional accuracy)
- RMSE (error magnitude)
- Quarterly bias breakdown (Q1–Q4 performance by season)

Business Insight

Enables targeted planner overrides: Instead of blanket adjustments, fix only the under/over-targeted stores with their quarterly bias breakdown.

Role-of-Store Segmentation: 5 Segments

Derived from model outputs + market features (CAGR HH, housing_new_share, total_competitor_ta)

High Growth	Rising demand, low competition	**Invest, expand**
Growth	Moderate growth trajectory	**Maintain investment**
Neutral	Stable, well-calibrated predictions	**Optimize operations**
Maintain	Mature stores, stable forecast	**Efficiency focus**
Defend	High competition, declining demand	**Tactical response**

Interactive Streamlit Demo App

1. Dataset Summary — 269K rows, 1,727 stores
2. Phase 1 Features — 35 features in 5 families
3. Model Configuration — XGBoost, LightGBM, Ridge
4. Model Results — WAPE, Bias, R^2 comparison
5. Store Accuracy Loop — per-store classification
6. Role-of-Store — 5 segments, business actions
7. AI Commentary — Gemini 2.0 Flash insights
8. Store Drill-Down — quarterly bias breakdown
9. Download Results — export CSVs
10. Upload & Score — predictions on new data

Live Application

https://storewisetargets.streamlit.app/

Technical Architecture

Python Modules

`src/feature_engineering.py`	build_phase1_features() — 40 features, leakage guard
`src/train_model.py`	Full pipeline: load → engineer → split → train
`src/store_accuracy_loop.py`	Per-store metrics + Role-of-Store assignment
`app/streamlit_app.py`	Interactive demo — 10 sections, 749 lines

Deployment

- Streamlit Cloud: https://storewisetargets.streamlit.app/
- Git integration: Auto-deploy from GitHub on push
- Model artifacts: XGBoost pickle in repo
- API integration: Gemini 2.0 Flash for AI commentary

Key Findings & Recommendations

Key Findings

- Lag features are power drivers (corr 0.91–0.98)
- XGBoost 7.2% WAPE vs 8.9% naive baseline (19% improvement)
- Store accuracy varies: 40% well-calibrated, 30% under, 30% over-targeted
- Quarterly bias reveals seasonality: Adjust Q4 for holiday demand

Recommendations

- Deploy XGBoost for Q3/Q4 planning; monitor WAPE weekly
- Use Role-of-Store segmentation for targeted overrides
- Quarterly re-scoring: Retrain on rolling 2-year window
- Phase 2: Add price elasticity, promotions, traffic data

Thank You

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Questions?

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